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Studierichting: Informatica

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$$\frac{\log_4 x}{\log_4 3} \cdot \log_3 \frac{x}{9} = 3 \cdot 27^{\log_3 2}$$

$$\Rightarrow \log_3 x \cdot (\log_3 x - 2) = 3 \cdot 3^{3 \log_3 2}$$

$$\text{Neem } t = \log_3 x$$

$$\Rightarrow t^2 - 2t = 3^{\log_3 8} \cdot 3$$

$$\Rightarrow t^2 - 2t - 24 = 0$$

$$\text{Dan } s = -\frac{b}{a} = +2 \text{ en } p = \frac{c}{a} = -24$$

$$\Rightarrow t_1 = 6 \text{ en } t_2 = -4$$

$$\Rightarrow x_1 = 3^6 = 729 \text{ en } x_2 = 3^{-4} = \frac{1}{81}$$

References